

Multi-Orbit Multibeam Satellite Soft Handover Strategy Based on Rate-Splitting Multiple Access

Shuai Han^{ID}, Senior Member, IEEE, Zhiqiang Li^{ID}, Graduate Student Member, IEEE,
Weixiao Meng^{ID}, Senior Member, IEEE, and Cheng Li^{ID}, Senior Member, IEEE

Abstract—Satellite communication technology has rapidly developed to provide global information services, where low-Earth-orbit (LEO) satellites are the most popular due to lower transmission delay and easier deployment compared to geosynchronous-orbit (GEO) satellites. However, LEO satellites provide information services for a short time due to rapid movement relative to the Earth. When there are no visible LEO satellites, communication will be interrupted, and users must frequently detect accessible satellites, which wastes transmission power and decreases communication quality of service (QoS). To provide continuous information services and reduce detection frequency, we propose a GEO and LEO satellite joint service scheme to improve the communication QoS during the handover process between LEO satellites. Considering the access flexibility and spectrum efficiency, we further introduce a rate-splitting multiple access for the proposed multi-orbit satellite joint service scheme. We establish corresponding optimization problems for different communication scenarios using the weighted sum rate and max-min information rate as measurement indicators. To solve these non-convex optimization problems, we propose different alternating optimization algorithms to transform the initial problems into alternating convex problems. The simulation results show that our design handover schemes improve the communication QoS and reduce detection frequency, indirectly improving energy and spectrum efficiency.

Index Terms—Multibeam satellite communication, satellite soft handover, interference management, rate-splitting strategy.

I. INTRODUCTION

SEAMLESS information services is an essential ability of the future communication networks [1], [2]. In areas where terrestrial networks are unable to cover or have limited terrestrial network resources, satellite communication networks can provide multimedia services for fixed and mobile users in these areas, which has played an increasingly important role in the future integrated network [3], [4].

A. Related Works

As typical resource-constrained systems, satellite communication systems widely adopt multibeam technology due

to the high spectrum efficiency (SE), where each satellite beam reuses spectrum resources (SRs) [5], [6]. Based on the geosynchronous-orbit (GEO) multibeam satellite communication systems (MSCSs), some broadcast, multicast, and unicast information services have been explored [7], which fully utilize the more extensive coverage of GEO MSCSs. However, the orbit altitude of the GEO satellite is 35786 km and is far from the ground, which makes it challenging to satisfy the various service requirements with the explosive growth of users. Meanwhile, the propagation latency from the terrestrial users to the GEO satellites is too long for some time-critical tasks [8]. In such cases, the low-Earth-orbit (LEO) satellites have attracted lots of interest with good development prospects. They are deployed on the low orbit at a height of 500-1500 km above the ground. It has the advantages of low propagation delay, low energy consumption, and suppression of signaling attenuation [9], which is conducive to real-time communication. Therefore, the LEO constellation will be widely applied in the near future [10].

Although LEO satellites have advantages such as low latency and easy deployment, mobility management of LEO satellites is more complex [11], [12]. Due to the rapid sliding relative to the Earth, the information service time provided by LEO satellites only continues for approximately ten minutes [13]. In such cases, users need to conduct frequent satellite measurements to ensure communication performance during handover. Therefore, how to design an effective handover mechanism is an urgent problem to be solved. The satellite handover contains two types, namely spot inter-beam and inter-satellite handovers [14]. The satellite handover can lead to many problems, such as delay, transmission loss, and signaling overhead. To reduce the number and complexity of handover, a shortest handover path algorithm is proposed [15], but it sacrifices the communication quality of service (QoS). Furthermore, a global aeronautical traffic demand map is proposed to overcome handovers between aircraft and LEO satellites using the multi-attribute decision-making algorithm [16]. As the constellation scale and the number of users increases, a network-flow model of the satellite handover is proposed to simplify the strategy [17].

As satellite communication technology develops, the number of users served by satellite increases rapidly, and the traditional orthogonal multiple access cannot support a massive number of users. Therefore, some non-orthogonal multiple access (NOMA) schemes based on multibeam satellites are proposed to weighted sum rate (WSR) [18] and

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Shuai Han, Zhiqiang Li, and Weixiao Meng are with the School of Electronics and Information Engineering, Harbin Institute of Technology, Harbin 150000, China (e-mail: hanshuai@hit.edu.cn; lizhiqiang0108@foxmail.com; wxmeng@hit.edu.cn).

Cheng Li is with the School of Engineering Science, Simon Fraser University, Burnaby, BC V5A 1S6, Canada (e-mail: cheng_li_5@sfu.ca).

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communication QoS [19]. However, the computational complexity will sharply increase as the number of users increases due to successive interference cancellation (SIC) algorithm [20]. Furthermore, much work has been created on NOMA in multi-antenna wireless networks [21]. However, NOMA is not efficient in multi-antenna settings [22]. That is because the NOMA literature, initially developed for single-antenna setups, has been hastily applied to multi-antenna setups, resulting in a misuse of spatial dimensions [22].

Recently, rate-splitting multiple access (RSMA) has emerged as a promising non-orthogonal technology, which has a flexible multiple access (MA) framework and robust interference management [23]. Therefore, many schemes based on RSMA have been designed for MSCSs to improve max-min rate (MMR) [24] and neutralize inter-beam interference [25]. Furthermore, some broadcast, multicast, and unicast joint serve schemes based on RSMA are proposed to meet the diversity information services of massive users [26], which fully utilizes the advantages of RSMA and MSCSs.

B. Motivation and Contribution

From the above literature analysis, many related works give excellent solutions. However, there are still many shortcomings in the existing works. First, the existing satellite handover strategies are based on the ultra-dense satellite-terrestrial integrated network, which optimizes resource allocation and satellite selection. However, the development of LEO satellites is at an early stage, and it is difficult to realize seamless global coverage. The existing strategies cannot guarantee communication QoS when there are no accessible LEO satellites. Furthermore, satellite systems are cooperative in the coexistence of high and low orbit satellites, and the existing work does not fully utilize the characteristics and advantages of different orbit satellites. Meanwhile, most of the existing handover schemes focus only on orthogonal resource allocation without optimizing specific physical layer MA technologies, which results in low access density and SE. Therefore, we propose a multi-orbit multibeam satellite soft handover scheme based on RSMA to make up for these shortcomings of the existing works. The main contributions are as follows:

- Based on the coverage range and distance differences of GEO and LEO satellites, we establish a joint soft-handover model, which fully guarantees the communication QoS during the handover between LEO satellites and provides users with high-quality continuous information services. Meanwhile, due to the introduction of GEO satellites with a large coverage area, when users cannot connect to LEO satellites, they can still obtain information services through GEO satellites to avoid communication interruption. This also avoids frequently detecting accessible LEO satellites by users when the current LEO satellite in service is invisible.
- The promising RSMA can simultaneously utilize the non-orthogonal transmission and precoding strategies by dividing the user messages into common and private sub-messages, which have high access flexibility and anti-interference ability [23], [27]. Therefore, we introduce rate-splitting (RS) strategy into the GEO and LEO

satellite joint soft-handover scheme, improving user communication QoS and access density.

- For two scenarios, i.e., emphasizing the overall serviceability and the information service quality of each user, we introduce two kinds of indicators, WSR and MMR. Then, we design the corresponding schemes and optimization problems for different indicators considering user communication QoS.
- To address these established non-convex optimization problems, we propose two alternating optimization algorithms to transform these non-convex problems into convex ones based on Lagrange dual transform and weighted minimum mean square error (WMMSE). Simulation results show that our design soft handover scheme based on RSMA can fully guarantee the continuity of user information service and improve the information transmission rate when LEO satellites move rapidly relative to the Earth, which also indirectly improves the energy efficiency (EE) and SE.

C. Organization and Notations

The structure of this paper is as follows. Section II presents the system model, which includes both the channel and signal models for MSCSs. Section III formulates the optimization problem for maximizing the MMR in a joint GEO and LEO satellite handover scheme. Taking into account the overall system's information serviceability, Section IV formulates the optimization problem for the WSR in GEO and LEO MSCSs and proposes a WMMSE-based AO algorithm to solve the non-convex WSR problem. In Section V, simulation results demonstrate the advantages of the proposed RSMA-based joint soft handover scheme for GEO and LEO satellites. Finally, Section VI concludes the paper.

Notations: Boldface uppercase letters denote matrices, boldface lowercase letters denote column vectors, and standard letters denote scalars. The superscripts $(\cdot)^*$, $(\cdot)^T$ and $(\cdot)^H$ denote conjugate, transpose and conjugate-transpose (Hermitian) operators, respectively. $\text{tr}(\cdot)$ and $\text{diag}(\cdot)$ are the trace and diagonal entries respectively. $\mathbb{E}(\cdot)$ denotes the expectation. The set of all N_1 -by- N_2 complex (real) matrices is denoted as $\mathbb{C}^{N_1 \times N_2}$ ($\mathbb{R}^{N_1 \times N_2}$). $\Re(\cdot)$ is the real part of a complex number, and $|\cdot|$ is the absolute value of variable.

II. SYSTEM MODEL

We consider a GEO and LEO satellite joint service system, as shown in Fig. 1,¹ where the GEO satellite has extensive coverage and can provide seamless information services to users. In contrast, LEO satellites cover a small area and move fast relative to the Earth. However, the distance between

¹Despite numerous challenges, such as communication delays, orbital dynamics, and interference management, multi-orbit satellite joint service solutions prove effective due to strategic implementations. These include advanced resource management algorithms, dynamic beamforming, efficient satellite switching technologies, and robust link designs. Additionally, comprehensive system-level testing ensures these technologies perform reliably in complex environments. Such measures allow these systems to maintain extensive coverage, robustness, and adaptability, effectively overcoming the inherent challenges of satellite communications [28], [29].

LEO satellites and users is much shorter than between GEO satellites and users. Without loss generality, we assume the gateway is the control center to manage the multi-orbit MSCSs [30]. In this configuration, when a user is within the coverage of the LEO satellite beam, the LEO satellite primarily provides information services, while the GEO satellite assists with communication. Conversely, when the user falls outside the LEO satellite's coverage but remains within the GEO satellite's range, the GEO satellite assumes the primary role in providing information services. Therefore, this joint GEO and LEO satellite service model ensures continuous information services and enables soft handover between LEO satellites. Additionally, users do not need to frequently detect accessible LEO satellites, as the gateway informs them in advance of the next available LEO satellite. In Fig. 1, θ represents the deviation angle, indicating how much the user deviates from the center of the current LEO satellite beam. Since GEO satellites are relatively geostationary, and the user's mobility speed is much slower than that of the satellites, we neglect any change in the deviation angle for the GEO satellite beam.

There are two handover cases for users: one is to switch between different beams of the same satellite, and the other is to switch between beams of different satellites. The former only requires one satellite to adjust user access, which can easily ensure communication QoS during the handover process. The latter requires all satellite systems to cooperate, and the handover process is more complex, which significantly impacts the quality of information services. Therefore, we only analyze the second case in our paper. For the convenience of analysis, we assume each satellite provides one satellite beam in the GEO and LEO satellite joint service system.

A. Channel Model

Satellite channel mainly considers free space propagation, shadow fading, and small-scale fading.² Without loss of generality, the channel is assumed to be flat fading and known at the transmitter and receiver,³ where doppler frequency shifts caused by LEO satellite is compensated using proper estimation techniques [33], [34], [35]. Meanwhile, the transmission delays of GEO and LEO satellites can be compensated by users and satellites in advance [36], [37]. Therefore, the channel gain between user- k and N_L LEO satellites is $\mathbf{h}_k = [h_{1,k}, h_{2,k}, \dots, h_{N_L,k}] \in \mathbb{C}^{1 \times N_L}$, where $h_{n,k}, n \in N_L$ is

$$h_{n,k} = \sqrt{G_k B_{n,k} L_{n,k}} T_k, \quad (1)$$

where G_k denotes the receiving antenna gain of the user- k . $L_{n,k} = (c/4\pi f_c d_{n,k})^2$ is the free space loss [33], where $d_{n,k}$ is the distance between the LEO satellite- n and the user- k ,

²Since the impact of atmospheric absorption, tropospheric scintillation, rain attenuation can be negligible for frequencies below 6 GHz [31], we mainly analyze free space propagation, shadow fading, and small-scale fading.

³The channel state information (CSI) can be obtained by feedback/training sent from the users via a return channel, which already exists in DVB-S2. CSI at satellite may get perturbed owing to the propagation delay and atmospheric dynamics. In fact, the CSI estimation procedure can be readily facilitated over a slow-varying fading channel [32].

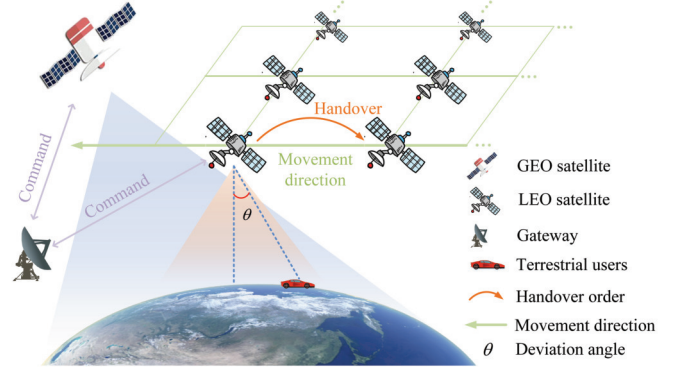


Fig. 1. The multi-orbit satellite joint service system.

f_c is the carrier frequency, c is the speed of light. $B_{n,k}, n \in N_L$ is radiation pattern of LEO satellite- n beam gain at user- k [38]

$$B_{n,k} = B_{\max} \left[\frac{J_1(u_{n,k})}{2u_{n,k}} + 36 \frac{J_3(u_{n,k})}{u_{n,k}^3} \right]^2, \quad (2)$$

where $u_{n,k} = 2.07123 \sin(\theta_{n,k})/\sin(\theta_{3dB})$. $\theta_{n,k}$ is the angle between user- k and the LEO satellite- n beam center, and θ_{3dB} is a 3 dB loss angle compared with the LEO satellite beam center. The maximum beam gain observed at each beam center is denoted by $B_{\max}$. J_1 and J_3 are respectively first-kind Bessel functions with order 1 and order 3. r_k is the shadowed-Rician fading [39], and the probability density function is

$$f_{r_k}(r) = \frac{ar}{b_0} \exp\left(-\frac{r^2}{2b_0}\right) {}_1F_1\left(m, 1, \frac{\Omega r^2}{2b_0(2b_0m + \Omega)}\right), \quad (3)$$

where $a = (2b_0m/2b_0m + \Omega)^m$, Ω and $2b_0$ are the average power of the line of sight and multi-path components, respectively, m is the fading severity parameter, and ${}_1F_1(\cdot; \cdot; \cdot)$ is the confluent hyper-geometric function of the first kind.

Similarly, the channel gain between user- k and N_G GEO satellites is $\mathbf{h}'_k = [h'_{1,k}, h'_{2,k}, \dots, h'_{N_G,k}] \in \mathbb{C}^{1 \times N_G}$, where $h'_{n,k}, n \in N_G$ is

$$h'_{n,k} = \sqrt{G_k B'_{n,k} L'_{n,k}} r_k, \quad (4)$$

where $L'_{n,k} = (c/4\pi f_c d'_{n,k})^2$ is the free space loss. The $d'_{n,k}$ is the distance between the GEO satellite- n and the user- k . $B'_{n,k}$ is radiation pattern of GEO satellite- n beam gain at user- k

B. Signal Models

We assume that K users share one carrier resource, where each user divides message W_k into a common message $W_{k,c}$ and a private message $W_{k,p}$ using 1-layer RS. The private messages $W_{1,p}, \dots, W_{K,p}$ are independently encoded into the private streams $s_1, \dots, s_K$ while the common messages $W_{1,c}, \dots, W_{K,c}$ are jointly encoded into a common stream s_0 . Then, linear precoding matrices $\mathbf{P} \in \mathbb{C}^{N_L \times (K+1)}$ at LEO satellites and $\mathbf{P}' \in \mathbb{C}^{N_G \times (K+1)}$ at GEO satellites are applied to the data stream vector $\mathbf{s} = [s_0, s_1, \dots, s_K] \in \mathbb{C}^{K+1}$, $\mathbb{E}(\mathbf{s}^H \mathbf{s}) = \mathbf{I}$. For the convenience of analysis, $\mathbf{P} = [\mathbf{q}_0, \mathbf{q}_1, \dots, \mathbf{q}_K] = [\mathbf{p}'_1, \dots, \mathbf{p}'_{N_L}]^T$, where $\mathbf{q}_0 = [p_{1,0}, p_{2,0}, \dots, p_{N_L,0}]^T$, $\mathbf{q}_k = [p_{1,k}, p_{2,k}, \dots, p_{N_L,k}]^T$, and $\mathbf{p}_n = [p_{n,0}, p_{n,1}, \dots, p_{n,K}]$.

Similarly, $\mathbf{P}' = [\mathbf{q}'_0, \mathbf{q}'_1, \dots, \mathbf{q}'_K] = [\mathbf{p}'_1^T, \dots, \mathbf{p}'_{N_G}^T]^T$, where $\mathbf{q}'_0 = [p'_{1,0}, p'_{2,0}, \dots, p'_{N_G,0}]^T$, $\mathbf{q}'_k = [p'_{1,k}, p'_{2,k}, \dots, p'_{N_G,k}]^T$, and $\mathbf{p}'_n = [p'_{n,0}, p'_{n,1}, \dots, p'_{n,K}]^T$. Then, the transmitted signals of LEO and GEO satellites are

$$x_n = \mathbf{p}_n \mathbf{s}^T = \underbrace{p_{n,0} s_0}_{\text{common data}} + \underbrace{\sum_{k \in K} p_{n,k} s_k}_{\text{private data}}, n \in N_L, \quad (5a)$$

$$x'_l = \mathbf{p}'_l \mathbf{s}^T = \underbrace{p'_{l,0} s_0}_{\text{common data}} + \underbrace{\sum_{k \in K} p'_{l,k} s_k}_{\text{private data}}, l \in N_G. \quad (5b)$$

Based on the joint service of multi-orbit satellites, the received signal y_k at user- k is

$$y_k = \sum_{n \in N_L} h_{n,k} x_n + \sum_{l \in N_G} h'_{l,k} x'_l + n_k, \quad (6)$$

where n_k is the additive white Gaussian noise (AWGN) with zero mean and variance σ_k^2 . We specify that a data stream intended for more users has a higher decoding priority [40]. Therefore, the common data will be first decoded, and the corresponding signal-to-interference plus noise ratio (SINR) of decoding s_0 is

$$\gamma_{k,0} = \frac{|\mathbf{h}_k \mathbf{q}_0 + \mathbf{h}'_k \mathbf{q}'_0|^2}{\sum_{j \in K} |\mathbf{h}_k \mathbf{q}_j + \mathbf{h}'_k \mathbf{q}'_j|^2 + \sigma_k^2}. \quad (7)$$

After decoding s_0 and canceling it from the received signal y_k using SIC, user- k decodes the private data s_k by treating the other private data as noise. Therefore, the corresponding SINR of decoding s_k is

$$\gamma_k = \frac{|\mathbf{h}_k \mathbf{q}_k + \mathbf{h}'_k \mathbf{q}'_k|^2}{\sum_{j \in K, j \neq k} |\mathbf{h}_k \mathbf{q}_j + \mathbf{h}'_k \mathbf{q}'_j|^2 + \sigma_k^2}. \quad (8)$$

According to the signal analysis, the corresponding achievable rates of decoding s_0 and s_k at user- k are $R_{k,0} = \log_2(1 + \gamma_{k,0})$ and $R_k = \log_2(1 + \gamma_k)$. Note that the actual information transmission rate R_0 of decoding s_0 should not exceed the $R_{k,0}, \forall k \in K$, which ensures that all users can decode s_0 successfully. Because R_0 is the achievable rate shared by K users, it can be expressed as $R_0 = \sum_{k \in K} C_k$, where C_k is the portion of R_0 corresponding to user- k . Then, the achieving rate of the user- k at user- k is

$$R_{k,\text{tot}} = C_k + R_k, \quad \forall k \in K. \quad (9)$$

III. MMR OPTIMIZATION PROBLEM FOR THE GEO AND LEO JOINT SOFT HANDOVER SCHEME BASED ON RSMA

Then, based on the signal analysis of users, the optimization problem of MMR maximization for users is formulated as

$$\max_{\mathbf{P}, \mathbf{P}', \mathbf{c}} \min_{k \in K} R_{k,\text{tot}} \quad (10a)$$

$$\text{s.t.} \quad \sum_{j \in K} C_j \leq R_{k,0}, \quad \forall k \in K \quad (10b)$$

$$C_k \geq 0, \quad \forall k \in K \quad (10c)$$

$$|\mathbf{p}_n|^2 \leq P_t, \quad \forall n \in N_L \quad (10d)$$

$$|\mathbf{p}'_l|^2 \leq P'_t, \quad \forall l \in N_G \quad (10e)$$

where constraint (10b) guarantees that all users can decode the common data successfully, and $\mathbf{c}$ is vector $[C_1, \dots, C_K]$. Constraint (10c) ensures that the common data rate for each user is not negative. Constraints (10d) and (10e) ensure that the transmit power allocated to one carrier by LEO and GEO satellites will not exceed the maximum transmit power P_t and P'_t , respectively.

Obviously, the objective function (10a) and constraint (10b) are both non-convex, which makes it difficult to solve the Problem (10) directly. Therefore, we introduce the Lagrangian dual transform based on the fractional programming technique [41], [42]. For the non-convex objective function, we build a new function

$$f_k(\mathbf{P}, \mathbf{P}', \mathbf{c}, \boldsymbol{\alpha}_2) = \log(1 + \alpha_k) - \alpha_k + \frac{(1 + \alpha_k) \gamma_k}{1 + \gamma_k} + C_k, k \in K, \quad (11)$$

where $\boldsymbol{\alpha}_2$ is auxiliary variables for the decoding SINRs γ_k , which refers to $[\alpha_1, \dots, \alpha_k, \dots, \alpha_K]^T$. Similarly, we build the new function for the non-convex constraint (10b)

$$f_{k,0}(\mathbf{P}, \mathbf{P}', \boldsymbol{\alpha}_1) = \log(1 + \alpha_{k,0}) - \alpha_{k,0} + \frac{(1 + \alpha_{k,0}) \gamma_{k,0}}{1 + \gamma_{k,0}}, k \in K, \quad (12)$$

where $\boldsymbol{\alpha}_1$ is auxiliary variables for the decoding SINRs $\gamma_{k,0}$, which refers to $[\alpha_{0,1}, \dots, \alpha_{0,k}, \dots, \alpha_{0,K}]^T$. Then, the Problem (10) can be equivalently reformulated as

$$\max_{\mathbf{P}, \mathbf{P}', \mathbf{c}, \boldsymbol{\alpha}_1, \boldsymbol{\alpha}_2} \min_{k \in K} f_k(\mathbf{P}, \mathbf{P}', \mathbf{c}, \boldsymbol{\alpha}_2) \quad (13a)$$

$$\text{s.t.} \quad \sum_{j \in K} C_j \leq f_{k,0}(\mathbf{P}, \mathbf{P}', \boldsymbol{\alpha}_1), \quad \forall k \in K \quad (13b)$$

(10c), (10d), (10e),

which can be solved by alternating algorithm, namely the points $(\mathbf{P}, \mathbf{P}', \mathbf{c})$ and $(\boldsymbol{\alpha}_1, \boldsymbol{\alpha}_2)$ are optimized alternatively. When $(\mathbf{P}, \mathbf{P}', \mathbf{c})$ are fixed, the optimal auxiliary variables $\alpha_{k,0}^{\text{opt}} = \gamma_{k,0}$ and $\alpha_k^{\text{opt}} = \gamma_k$ can be obtained at $\partial f_{k,0} / \partial \alpha_{k,0} = 0$ and $\partial f_k / \partial \alpha_k = 0$, respectively.

Algorithm 1 MMR Maximization Based on Lagrange Dual Transform

Input: $N_L, N_G, P_t, P'_t, G_k, \theta_{\text{3dB}}, \theta'_{\text{3dB}}$.

- 1: **Initialization:** $i \leftarrow 0, \mathbf{P}^{[i]}, \mathbf{P}'^{[i]}, \text{MMR}^{[i]}$
- 2: **while** $|\text{MMR}^{[i]} - \text{MMR}^{[i-1]}| \geq \delta$ **do**
- 3: $i \leftarrow i + 1$;
- 4: Update the nominal SINR $\alpha_1^{[i]}$ and $\alpha_2^{[i]}$;
- 5: Update $\beta_1^{[i]}$ and $\beta_2^{[i]}$;
- 6: Update transmit beamforming $\mathbf{P}^{[i]}, \mathbf{P}'^{[i]}$, and $\mathbf{c}^{[i]}$;
- 7: **end while**
- 8: **Outputs:** $(\mathbf{P}^{\text{opt}}, \mathbf{P}'^{\text{opt}}, \mathbf{c}^{\text{opt}}), (\beta_1^{\text{opt}}, \beta_2^{\text{opt}}), (\alpha_1^{\text{opt}}, \alpha_2^{\text{opt}})$, and MMR

Furthermore, we fix the auxiliary variables $\boldsymbol{\alpha}_1$ and $\boldsymbol{\alpha}_2$, and then optimize $(\mathbf{P}, \mathbf{P}', \mathbf{c})$

$$\max_{\mathbf{P}, \mathbf{P}', \mathbf{c}} \min_{k \in K} f_k(\mathbf{P}, \mathbf{P}', \mathbf{c}, \boldsymbol{\alpha}_2) \quad (14a)$$

$$(10c), (10d), (10e), (13b),$$

Since $\frac{(1+\alpha_k)\gamma_k}{1+\gamma_k}$ in (14a) and $\frac{(1+\alpha_{k,0})\gamma_{k,0}}{1+\gamma_{k,0}}$ in (13b) are still non-convex in Problem (14), we further introduce quadratic transform proposed in [42] to solve the multiple-ratio fractional programming problem, namely

$$f_{k,0}^1(\mathbf{P}, \mathbf{P}', \beta_1) = 2\sqrt{1+\alpha_{k,0}}\text{Re}\{\beta_{k,0}^*(\mathbf{h}_k\mathbf{q}_0 + \mathbf{h}'_k\mathbf{q}'_0)\} - |\beta_{k,0}|^2 \left(\sum_{i=1}^K |\mathbf{h}_k\mathbf{q}_i + \mathbf{h}'_k\mathbf{q}'_i|^2 + \sigma_k^2 \right), \quad (15)$$

$$f_k^1(\mathbf{P}, \mathbf{P}', \beta_2) = 2\sqrt{1+\alpha_k}\text{Re}\{\beta_k^*(\mathbf{h}_k\mathbf{q}_k + \mathbf{h}'_k\mathbf{q}'_k)\} - |\beta_k|^2 \left(\sum_{i=1, i \neq k}^K |\mathbf{h}_k\mathbf{q}_i + \mathbf{h}'_k\mathbf{q}'_i|^2 + \sigma_k^2 \right), \quad (16)$$

where the fractional-programming auxiliary variables β_1 and β_2 are $[\beta_1, \dots, \beta_{k,0}, \dots, \beta_K, 0]^T$ and $[\beta_1, \dots, \beta_k, \dots, \beta_K]^T$, respectively. Problem (14) can be equivalently rewritten as

$$\max_{\mathbf{P}, \mathbf{P}', \mathbf{c}, \beta_1, \beta_2} \min_{k \in K} f_k^2(\mathbf{P}, \mathbf{P}', \mathbf{c}, \alpha_2) \quad (17a)$$

$$\text{s.t.} \quad \sum_{j \in K} C_j \leq f_{k,0}^2(\mathbf{P}, \mathbf{P}', \alpha_1), \quad \forall k \in K \quad (17b)$$

(10c), (10d), (10e),

where $f_k^2, k \in K$ and $f_{k,0}^2, k \in K$ are

$$f_k^2(\mathbf{P}, \mathbf{P}', \mathbf{c}, \alpha_2, \beta_2) = \log(1 + \alpha_k) - \alpha_k + f_k^1 + C_k, k \in K, \quad (18a)$$

$$f_{k,0}^2(\mathbf{P}, \mathbf{P}', \alpha_1, \beta_1) = \log(1 + \alpha_{k,0}) - \alpha_{k,0} + f_{k,0}^1, k \in K. \quad (18b)$$

Then, Problem (17) is convex problem when one of the two blocks (β_1, β_2) and $(\mathbf{P}, \mathbf{P}', \mathbf{c})$ is fixed. Therefore, Problem (17) can be updated alternatively. When we fix $(\mathbf{P}, \mathbf{P}', \mathbf{c})$, the optimal fractional-programming auxiliary variables $(\beta_1^{\text{opt}}, \beta_2^{\text{opt}})$ are given by setting $\partial f_2 / \partial \beta_k$ to zero [41], [42],

$$\beta_{k,0}^{\text{opt}} = \frac{\sqrt{1+\alpha_{k,0}}(\mathbf{h}_k\mathbf{q}_0 + \mathbf{h}'_k\mathbf{q}'_0)}{\sum_{i=1}^K |\mathbf{h}_k\mathbf{q}_i + \mathbf{h}'_k\mathbf{q}'_i|^2 + \sigma_k^2}, \quad (19a)$$

$$\beta_k^{\text{opt}} = \frac{\sqrt{1+\alpha_k}(\mathbf{h}_k\mathbf{q}_k + \mathbf{h}'_k\mathbf{q}'_k)}{\sum_{i=1, i \neq k}^K |\mathbf{h}_k\mathbf{q}_i + \mathbf{h}'_k\mathbf{q}'_i|^2 + \sigma_k^2}. \quad (19b)$$

Furthermore, when we fix (β_1, β_2) , the Problem (17) will become a second-order cone program (SOCP) problem, and then be solved by convex optimization method. Therefore, the initial optimization Problem (10) can be transformed into Problem (20)

$$\max_{\mathbf{P}, \mathbf{P}', \mathbf{c}, \alpha_1, \alpha_2, \beta_1, \beta_2} \min_{k \in K} f_k^2(\mathbf{P}, \mathbf{P}', \mathbf{c}, \alpha_2, \beta_2) \quad (20a)$$

s.t. (10c), (10d), (10e), (17b),

where the three blocks $(\mathbf{P}, \mathbf{P}', \mathbf{c})$, (α_1, α_2) , and (β_1, β_2) are alternatively optimized when the two other blocks are fixed. The detailed optimization steps are described in Algorithm 1.

In Algorithm 1, MMR^[2] is increasing and bounded with given transmit power because Problem (20) is a

convex problem in each block $(\mathbf{P}, \mathbf{P}', \mathbf{c})$, (β_1, β_2) , and (α_1, α_2) when the two other blocks are fixed [41], [42]. Therefore, the Algorithm 1 is converge. The computational complexity of updating (β_1, β_2) , and (α_1, α_2) are $\mathcal{O}(K^2 N_L + K^2 N_G)$. The $(\mathbf{P}, \mathbf{P}', \mathbf{c})$ are updated by solving the second-order cone program (SOCP) problem with computational complexity $\mathcal{O}((K N_L + K N_G)^{3.5})$. Then, the computational complexity of Algorithm 1 is approximately equal to $\mathcal{O}((K N_L + K N_G)^{3.5} \log(\delta^{-1}))$, where δ^{-1} is the convergence tolerance of the algorithm.

IV. WSR OPTIMIZATION PROBLEM FOR THE GEO AND LEO JOINT SOFT HANDOVER SCHEME BASED ON RSMA

This section analyzes the WSR indicator, which measures the overall service capability. Based on the signal analysis of users, the optimization problem of WSR maximization for users is formulated as

$$\max_{\mathbf{P}, \mathbf{P}', \mathbf{c}} \sum_{k \in K} R_{k,\text{tot}} \quad (21a)$$

$$\text{s.t.} \quad C_k + R_k \geq R_k^{\text{th}}, \quad \forall k \in K \quad (21b)$$

(10c), (10d), (10e),

where constraint (21b) ensures the QoS rate of each user.

To solve the non-convex problem (21), we introduce an improved AO algorithm based on WMMSE to transform the non-convex problem into an alternating convex problem. We specify that s_0 and s_k are detected using the equalizers $g_{k,0}$ and g_k , respectively. At user- k , $\hat{s}_{k,0}$ is the estimation of s_0 , i.e., $\hat{s}_{k,0} = g_{k,0}y_k$, and then cancel it from the received signal y_k , we can further obtain the estimation of s_k , i.e., $\hat{s}_k = g_k \left[y_k - \left(\sum_{n \in N_L} h_{n,k} p_{n,0} + \sum_{l \in N_G} h'_{l,k} p'_{l,0} \right) \hat{s}_0 \right]$. Therefore, we first calculate the mean square errors of demodulating s_0 and s_k

$$\begin{aligned} \varepsilon_{k,0} &\triangleq \mathbb{E} \{ |\hat{s}_{k,0} - s_0|^2 \} \\ &= |g_{k,0}|^2 T_{k,0} \\ &\quad - 2\Re \left(\sum_{n \in N_L} g_{k,0} h_{n,k} p_{n,0} + \sum_{l \in N_G} g_{k,0} h'_{l,k} p'_{l,0} \right) + 1, \end{aligned} \quad (22a)$$

$$\begin{aligned} \varepsilon_k &\triangleq \mathbb{E} \{ |\hat{s}_k - s_k|^2 \} \\ &= |g_k|^2 T_k \\ &\quad - 2\Re \left(\sum_{n \in N_L} g_k h_{n,k} p_{n,k} + \sum_{l \in N_G} g_k h'_{l,k} p'_{l,k} \right) + 1, \end{aligned} \quad (22b)$$

where $T_{k,0} = |\sum_{n \in N_L} h_{n,k} p_{n,0} + \sum_{l \in N_G} h'_{l,k} p'_{l,0}|^2 + \sum_{j \in K} |\sum_{n \in N_L} h_{n,k} p_{n,k} + \sum_{l \in N_G} h'_{l,k} p'_{l,k}|^2 + \sigma_{n,k}^2$ and $T_k = T_{k,0} - |\sum_{n \in N_L} h_{n,k} p_{n,0} + \sum_{l \in N_G} h'_{l,k} p'_{l,0}|^2$. When $\frac{\partial \varepsilon_{k,0}}{\partial g_{k,0}} = 0$ and $\frac{\partial \varepsilon_k}{\partial g_k} = 0$, then we can obtain the optimal equalizers

$$g_{k,0}^{\text{MMSE}} = \left(\sum_{n \in N_L} p_{n,0}^H h_{n,k}^H + \sum_{l \in N_G} p'_{l,0}^H h'_{l,k}^H \right) T_{k,0}^{-1}, \quad (23a)$$

$$g_k^{\text{MMSE}} = \left(\sum_{n \in N_L} p_{n,k}^H h_{n,k}^H + \sum_{l \in N_G} p'_{l,k}^H h'_{l,k}^H \right) T_k^{-1}. \quad (23b)$$

Substituting (23) into (22), the MMSEs can be obtained

$$\varepsilon_{k,0}^{\text{MMSE}} = 1 - T_{k,0}^{-1} \left| \sum_{n \in N_L} h_{n,k} p_{n,0} + \sum_{l \in N_G} h'_{l,k} p'_{l,0} \right|^2, \quad (24a)$$

$$\varepsilon_k^{\text{MMSE}} = 1 - T_k^{-1} \left| \sum_{n \in N_L} h_{n,k} p_{n,k} + \sum_{l \in N_G} h'_{l,k} p'_{l,k} \right|^2. \quad (24b)$$

Then, the corresponding SINRs can be rewritten as $\gamma_{k,0} = \frac{1}{\varepsilon_{k,0}^{\text{MMSE}}} - 1$ and $\gamma_k = \frac{1}{\varepsilon_k^{\text{MMSE}}} - 1$. Building upon this, the relationships between the information rates and MSEs can be expressed as $R_{k,0} = -\log_2(\varepsilon_{k,0}^{\text{MMSE}})$ and $R_k = -\log_2(\varepsilon_k^{\text{MMSE}})$. Furthermore, we introduce positive weights $(v_{k,0}, v_k)$ and define $\zeta_{k,0}$ and ζ_k [40]

$$\zeta_{k,0} \triangleq v_{k,0} \varepsilon_{k,0} - \log_2(v_{k,0}), \quad (25a)$$

$$\zeta_k \triangleq v_k \varepsilon_k - \log_2(v_k). \quad (25b)$$

Then, the rate-WMMSE relationships are established as

$$\zeta_{k,0}^{\text{MMSE}} \triangleq \min_{v_{k,0}, g_{k,0}} \zeta_{k,0} = 1 - R_{k,0}, \quad (26a)$$

$$\zeta_k^{\text{MMSE}} \triangleq \min_{v_k, g_k} \zeta_k = 1 - R_k, \quad (26b)$$

where the optimal $\zeta_{k,0}^{\text{MMSE}}$ and ζ_k^{MMSE} can be obtained at points $g_{k,0}^{\text{MMSE}}, g_k^{\text{MMSE}}, v_{k,0}^{\text{MMSE}} = (\varepsilon_{k,0}^{\text{MMSE}})^{-1}$, and $v_k^{\text{MMSE}} = (\varepsilon_k^{\text{MMSE}})^{-1}$, which can be verified by checking the first-order optimality conditions.

Algorithm 2 WSR Maximization Based on WMMSE

Input: $N_L, N_G, P_t, P'_t, G_k, \theta_{3\text{dB}}, \theta'_{3\text{dB}}$.

- 1: **Initialization:**
- 2: $i \leftarrow 0, \mathbf{P}^{[i]}, \mathbf{P}'^{[i]}, \text{WSR}^{[i]}$;
- 3: **while** $|\text{WSR}_{\text{tot}}^{[i]} - \text{WSR}_{\text{tot}}^{[i-1]}| \geq \delta$ **do**
- 4: $i \leftarrow i + 1$;
- 5: $\mathbf{P}^{[i-1]} \leftarrow \mathbf{P}^{[i]}, l \in N$;
- 6: $\mathbf{v} \leftarrow \mathbf{v}^{\text{MMSE}}, \mathbf{g} \leftarrow \mathbf{g}^{\text{MMSE}}$;
- 7: Update $\mathbf{P}, \mathbf{P}', \mathbf{x}$ with the updated $\mathbf{v}, \mathbf{g}$;
- 8: **end while**
- 9: **Outputs:** $\mathbf{P}^{\text{opt}}, \mathbf{P}'^{\text{opt}}, \mathbf{v}^{\text{opt}}, \mathbf{g}^{\text{opt}}, \mathbf{x}^{\text{opt}}, \text{WSR}_{\text{tot}}$

The problem (21) can be equivalently transformed into the alternating convex problem,

$$\min_{\mathbf{P}, \mathbf{P}', \mathbf{x}, \mathbf{v}, \mathbf{g}} \sum_{k \in K} \psi_{k,\text{tot}} \quad (27a)$$

$$\text{s.t.} \quad X_k + \zeta_k \leq 1 - R_k^{\text{th}}, \quad \forall k \in K \quad (27b)$$

$$\sum_{j \in K} X_j + 1 \geq \zeta_k, \quad \forall k \in K \quad (27c)$$

$$X_k \leq 0, \quad \forall k \in K \quad (27d)$$

$$(10d), (10e),$$

where $\mathbf{x} = [X_1, \dots, X_K]$ is the transformation of the common rate $\mathbf{c}$, i.e., $\mathbf{x} = -\mathbf{c}$. The MMSE weights and equalizers are $\mathbf{v} = [v_{1,0}, \dots, v_{K,0}, v_1, \dots, v_K]$ and $\mathbf{g} = [g_{1,0}, \dots, g_{K,0}, g_1, \dots, g_K]$, respectively. The $\psi_{k,\text{tot}}$ is defined as $\psi_{k,\text{tot}} = X_k + \zeta_k$. Let $\mathbf{v}^{\text{MMSE}}$ and $\mathbf{g}^{\text{MMSE}}$ denote the corresponding optimal MMSE equalizers and weights according to (26), which satisfies the Karush-Kuhn-Tucker conditions of problem (27) when $\mathbf{p}$ and $\mathbf{x}$ are fixed. Meanwhile, when the

TABLE I
SIMULATION PARAMETERS

Parameters	Value
GEO satellite orbit height	35786 Km
LEO satellite orbit height	1000 Km
Carrier frequency	2.5 GHz (L/S)
Subcarrier bandwidth	1 MHz
GEO satellite beam 3 dB angle	1.59°
LEO satellite beam 3 dB angle	5.7°
GEO Satellite maximum beam gain	52 dBi
LEO Satellite maximum beam gain	30 dBi

optimal points $\mathbf{p}^{\text{opt}}, \mathbf{x}^{\text{opt}}, \mathbf{v}^{\text{opt}}, \mathbf{g}^{\text{opt}}$ satisfy the problem (27), the points $\mathbf{p}^{\text{opt}}, \mathbf{c}^{\text{opt}} = -\mathbf{x}^{\text{opt}}$ satisfy the problem (21). Therefore, we can obtain the solutions to the problem (21) by solving the problem (27). Although the joint optimization of $(\mathbf{P}, \mathbf{P}', \mathbf{x})$ in problem (27) is still non-convex, it can become convex in each block $(\mathbf{P}, \mathbf{P}', \mathbf{x}), \mathbf{v}, \mathbf{g}$ when the other two blocks are fixed. The detailed optimization algorithm steps for users are described in Algorithm 2, where $(\mathbf{P}, \mathbf{P}', \mathbf{x}), \mathbf{v}$, and $\mathbf{g}$ are updated iteratively until the convergence of the WSR.

Since problem (27) can become convex in each block $(\mathbf{P}, \mathbf{x}), \mathbf{v}, \mathbf{g}$ when the other two blocks are fixed, $\text{WSR}^{[i]}$ is increasing at each iteration. Meanwhile, Algorithm 2 is bounded above for a given power constraint [43]. The three blocks are repeated alternating until convergence, as summarized in Algorithm 1. Therefore, Algorithm 2 is convergent. At each iteration, the computational complexity mainly depends on solving SOCP problem, which is $\mathcal{O}((KN_L + KN_G)^{3.5} \log(\delta^{-1}))$. Algorithm 1 and Algorithm 2 have the same order of magnitude computational complexity.

V. SIMULATION RESULTS

In this section, we analyze two scenarios, i.e., emphasizing the overall serviceability and user communication QoS, which corresponds to two indicators, WSR and MMR, respectively. Furthermore, we compare the proposed GEO and LEO satellite joint soft handover scheme based on RSMA with various baseline schemes from multiple perspectives. In the simulation, the shadowed-Rician fading is average shadowing ($b_0 = 0.126$, $m = 10.1$, $\Omega = 0.835$) [39]. We assume that antenna gains of users are randomly selected between 0 dB and 10 dB. The power of AWGN is $\sigma^2 = \kappa T_{\text{sys}} B$, where κ is the Boltzmann constant, and $T_{\text{sys}} = 290$ K is the system noise temperature in kelvin [31]. Without loss of generality, the transmit power P_t and P'_t are set to 10 W with different antenna gains, where some detailed parameters are listed in Table I [30], [31]. In the proposed GEO and LEO satellite joint soft handover system, each user obtains information service from one LEO satellite and one GEO satellite in a time slot. Therefore, the N_G and N_L are set to 1 in simulation.

A. Performance Analysis of MMR

For some communication scenarios, more attention is paid to communication QoS of each user. Therefore, we first discuss the MMR in the GEO and LEO joint service system.

When there is no GEO satellite assistance, we observe the change in MMR as the LEO satellite moves away in Fig. 2,

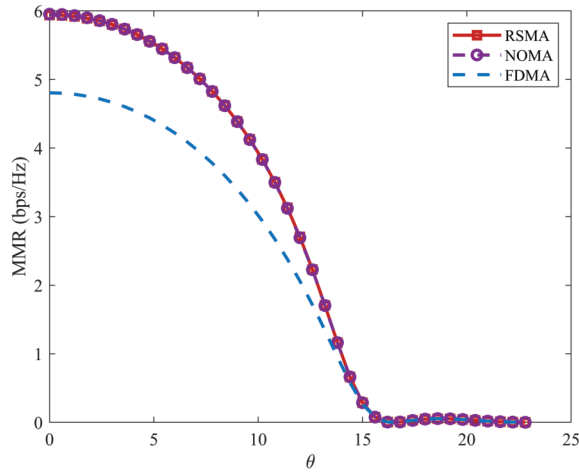


Fig. 2. The relationship between MMR and deviation angle θ under different access schemes with $K=2$ in the LEO satellite system, where the distance between adjacent LEO satellites is far more than 100 Km.

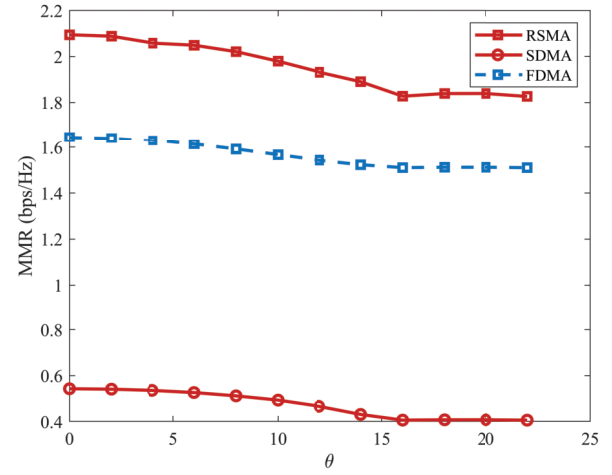


Fig. 4. The relationship between MMR and deviation angle θ of different access schemes at $K=4$ in GEO and LEO satellite joint service, where the distance between adjacent LEO satellites exceeds 1000 Km.

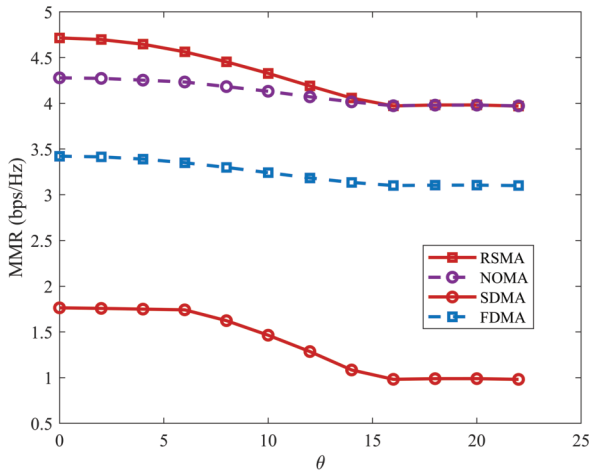


Fig. 3. The relationship between MMR and deviation angle θ of different access schemes at $K=2$ in GEO and LEO satellite joint service, where the distance between adjacent LEO satellites exceeds 1000 Km.

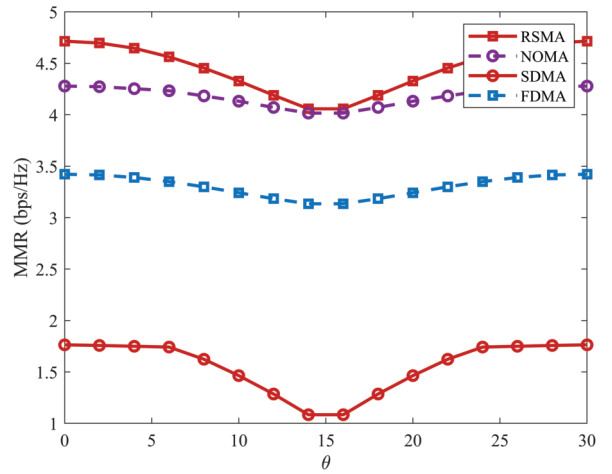


Fig. 5. The relationship between MMR and deviation angle θ of different access schemes at $K=2$ in GEO and LEO satellite joint service, where the distance between adjacent LEO satellites is 600 Km.

where the LEO satellite beam radius is 100 Km and K is set to 2. We can see that RSMA and NOMA have better MMR performance than FDMA schemes. The reason is that the system works in overload status for high access density. In such a case, RSMA and NOMA can utilize non-orthogonal transmission advantage. Meanwhile, we find that the MMRs of three MA modes decrease rapidly as the θ increases, namely, the LEO satellite moves away. Therefore, the proposed GEO and LEO satellite joint service scheme is required.

Continuing with the analysis in Fig. 2, we further discuss the performance of the proposed GEO and LEO satellite joint service scheme based on RSMA in Fig. 3, where the distance between adjacent LEO satellites exceeds 1000 Km, and the radius of GEO and LEO satellite beam are 1000 Km and 100 Km, respectively. In the GEO and LEO satellite joint service scheme, we see that the MMRs of all MA modes are improved significantly compared with Fig. 2 as the current LEO service satellite moves away. Meanwhile, we further observe that RSMA and FDMA have the best and worst

MMRs, respectively. The reason is that RSMA can utilize the non-orthogonal transmission and precoding strategies, and FDMA has low resource utilization efficiency. When the deviation angle θ is large enough, RSMA has the same MMR as NOMA in the same conditions. That is because only the GEO satellite provides information service when deviation angle θ is more than 15° . In such a case, RSMA will degenerate into the NOMA scheme. Furthermore, we improve the user access density and analyze the MMR in Fig. 4, where K is set to 4. It can be seen that with the assistance of GEO satellites, the MMR of users has significantly improved compared with Fig. 2. Compared with Fig. 3 at $K=2$, we observe the MMR decreases in the same condition in compared with Fig. 4. This is because the same transmit power will be allocated to more users.

To obtain more insight, we further discuss different distances between adjacent LEO satellites in Figs. 5 and 6, where the distance between adjacent LEO satellites is 600 kilometers. Unlike the phenomena in Figs. 3 and 4, we can see that MMRs

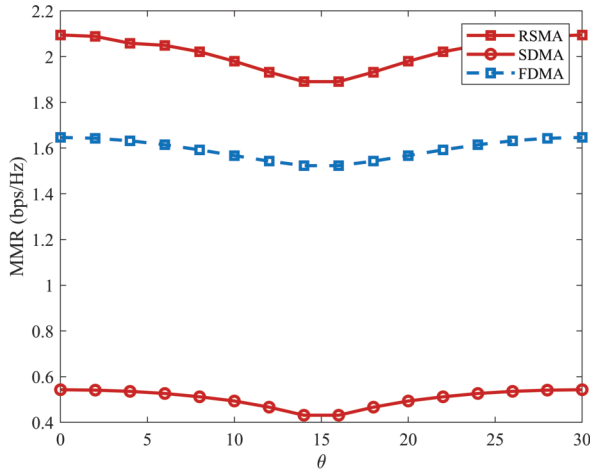


Fig. 6. The relationship between MMR and deviation angle θ of different access schemes at $K=4$ in GEO and LEO satellite joint service, where the distance between adjacent LEO satellites is 600 Km.

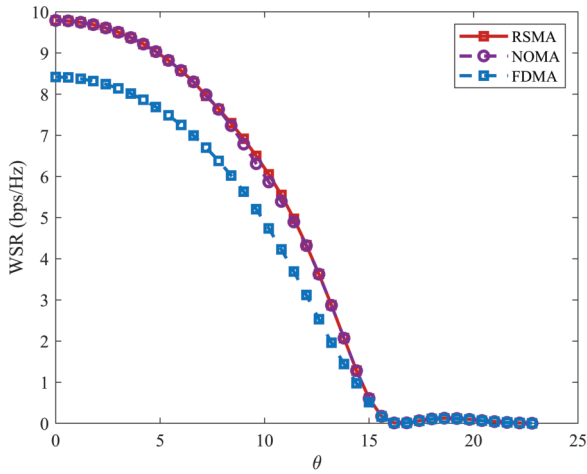


Fig. 7. The relationship between WSR and deviation angle θ under different access schemes with $K=2$ in the LEO satellite system, where the distance between adjacent LEO satellites is far more than 100 Km, R^{th} is set to 0.2 bps/Hz.

will first decrease and then increase in Figs. 5 and 6 as the θ change from 0° to 30° . The reason is that the deviation angle θ between the user and beam center of the current LEO satellite in service reaches 15° , the next LEO service satellite arrives, which shows that the proposed GEO and LEO satellite joint soft handover scheme can ensure the communication QoS during the handover process. Meanwhile, it can be seen that RSMA has the best performance. Compared with Fig. 5, the MMR of Fig. 6 is lower. The reasons have been explained above in detail.

B. Performance Analysis of WSR

In this part, we consider the second communication scenario, namely that the WSR is a concern when the lowest rate QoS is satisfied, which concerns the overall information service capacity.

In Fig. 7, we simulate the relationship between WSR and deviation angle θ with different schemes. Within the

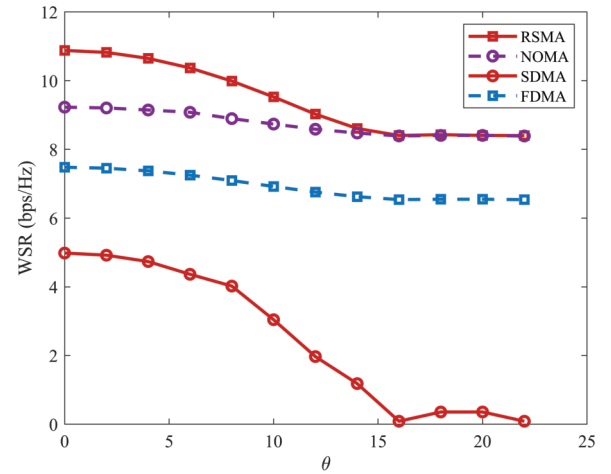


Fig. 8. The relationship between WSR and deviation angle θ of different access schemes at $K=2$ in GEO and LEO satellite joint service, where the distance between adjacent LEO satellites exceeds 1000 Km, R^{th} is set to 0.2 bps/Hz.

5.7° angle range, WSR slowly decreases as the angle increases. As the satellite continues to move away, and deviation angle θ exceeds 5.7° , the WSR performance rapidly decreases. Beyond this angle, as the satellite continues to move away, the performance rapidly deteriorates. This trend aligns with the satellite beam gain pattern, where at a $\theta = 10^\circ$ deviation angle, the WSR drops to an unacceptable value that does not meet the required user communication QoS. We can also see a slight increase in WSR at a 20° deviation angle, which is attributed to users being in the beam sidelobe range as the satellite moves away. The slight recovery in WSR at wider deviation angles suggests some residual capacity in the beam's sidelobes, but this is not enough to provide satisfactory service levels. Furthermore, compared to FDMA, NOMA and RSMA perform better in the same conditions because they utilize non-orthogonal characteristics. The evaluation in Fig. 7 illustrates the critical impact of satellite positioning and beam directionality on communication efficiency in satellite networks. It underscores the importance of our proposed multi-orbit satellite handover strategy, which maintains optimal alignment between the satellite beam and the user's location to ensure service quality.

In Fig. 8, we simulated the WSR changes of users as LEO satellites move away in the GEO and LEO satellite joint service schemes, where the distance between adjacent LEO satellites exceeds 1000 Km. From Fig. 8, we see that the WSR is significantly improved compared to Fig. 7 in the same conditions. This is because when the LEO satellite moves away, the GEO satellite plays a vital role in compensating for this loss. Meanwhile, the FDMA scheme has the lowest WSR because it cannot utilize a non-orthogonal transmission strategy. On the other hand, RSMA can simultaneously utilize non-orthogonal and precoding strategies, so it has the highest WSR. When deviation angle θ is large enough, the RSMA and NOMA schemes have the same WSRs. This is because LEO satellites hardly contribute information rate at that moment, and RSMA will degenerate into the NOMA because there is only one GEO satellite beam to serve users.

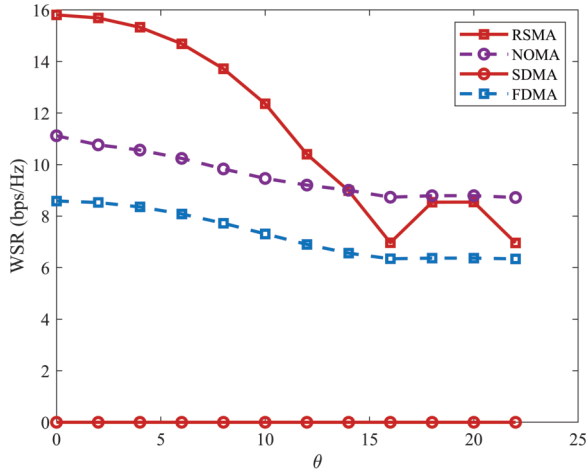


Fig. 9. The relationship between WSR and deviation angle θ of different access schemes at $K=4$ in GEO and LEO satellite joint service, where the distance between adjacent LEO satellites exceeds 1000 Km, R^{th} is set to 0.2 bps/Hz.

Furthermore, we set K to 4 for more user access density in Fig. 9. As the deviation angle θ increases, the WSR initially decreases rapidly, which aligns with the decrease in beam gain as users move away from the beam center of the LEO satellite. RSMA generally shows better performance compared to the other access schemes (like FDMA and NOMA) at lower deviation angles. This is because RSMA effectively manages non-orthogonal transmission and precoding, which provides higher flexibility and efficiency in managing interference. At certain angles (between 15° and 20°), NOMA's performance is shown to exceed RSMA. This happens because RSMA degenerates into NOMA when the LEO satellite is no longer a significant contributor, and the system relies primarily on the GEO satellite, which serves users using NOMA's 3-layer SIC. At this point, NOMA handles user interference better than RSMA's 1-layer SIC, resulting in better WSR for NOMA. FDMA consistently has the lowest WSR across the range of deviation angles due to its inefficient use of spectrum resources, as it relies on orthogonal transmission that limits access density and resource utilization. The simulation sets the number of users to 4, which increases the demand for system resources. As a result, there is more user interference, and schemes that handle interference better (such as RSMA and NOMA) show superior performance compared to FDMA. However, due to interference challenges, the WSR for RSMA slightly underperforms NOMA when the system relies heavily on the GEO satellite. RSMA generally performs best in managing interference and maximizing WSR when LEO satellites play a more active role in providing service.

Similar to the analysis of MMR, we further analyze the performance of WSR for GEO and LEO joint soft handover scheme based on RSMA technology, where the distance between adjacent LEO satellites is 600 Km. Without loss of generality, K and R^{th} are set to 2 and 0.2 bps/Hz, respectively. From Figs. 10 and 11, we find that WSRs are significantly improved during the movement of the LEO satellite compared with Fig. 7, which verifies the effectiveness of the proposed GEO and LEO joint soft handover scheme. As deviation angle

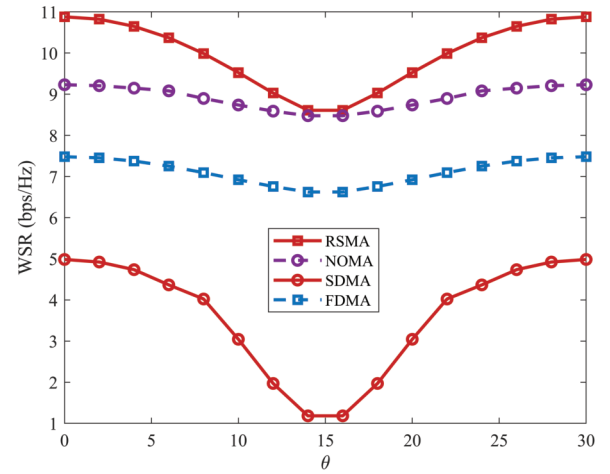


Fig. 10. The relationship between WSR and deviation angle θ of different access schemes at $K=2$ in GEO and LEO satellite joint service, where the distance between adjacent LEO satellites is 600 Km, R^{th} is set to 0.2 bps/Hz.

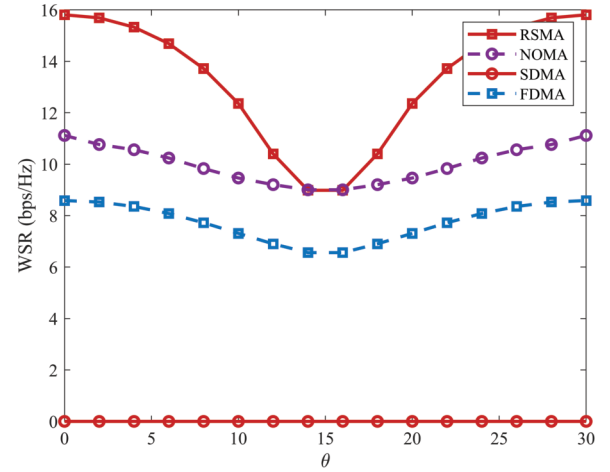


Fig. 11. The relationship between WSR and deviation angle θ of different access schemes at $K=4$ in GEO and LEO satellite joint service, where the distance between adjacent LEO satellites is 600 Km, R^{th} is set to 0.2 bps/Hz.

θ increases, WSR first decreases, but when it drops to a certain extent, it will continue to increase because the next LEO service satellite arrives. Therefore, our proposed scheme provides users with a high-quality information service during the whole handover process. Meanwhile, it can be seen that RSMA has consistently achieved the highest WSR compared with other MA modes in the same conditions. Compared with Fig. 10, the WSRs of RSMA, SDMA, and FDMA in Fig. 11 increase because more users access the GEO and LEO MSCS. Note that the WSR of SDMA is 0 bps/Hz in Fig. 11 since SDMA has serious MAI with $K = 4$, which makes it impossible for each user to obtain information rates greater than R^{th} .

In general, users only need to receive the switching command from GEO satellites without sending request signals when they switch between LEO satellites in the proposed GEO and LEO satellite joint soft handover scheme. This reduces the user handover time and transmit power. Since

the user is always in the GEO satellite coverage, there is no communication interruption. In such a case, the user does not need to frequently detect the available LEO satellites, which reduces the access delay and guarantees communication QoS during the handover process.

VI. CONCLUSION

In this study, we developed a joint soft handover model integrating GEO and LEO satellites, tailored to leverage the unique characteristics of satellites in different orbits. This model robustly maintains communication QoS during transitions between LEO satellites and significantly reduces the need for frequent satellite accessibility checks. We further enhanced this model by integrating RSMA technology, addressing two primary scenarios: optimizing overall system service capacity and improving individual user communication QoS. To quantitatively evaluate system performance, we introduced two metrics, WSR and MMR, and formulated corresponding optimization challenges. To solve these non-convex optimization issues, we devised two alternating optimization algorithms utilizing the Lagrangian dual transform and the WMMSE approach. Our simulation results affirm that the proposed GEO-LEO joint handover strategy, underpinned by RSMA, surpasses conventional approaches in enhancing SE, EE, and access density, thereby offering substantial improvements over baseline models. This underscores the efficacy of our proposed schemes in facilitating seamless satellite communication transitions, reinforcing the critical role of advanced algorithmic strategies in contemporary satellite communications.

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Shuai Han (Senior Member, IEEE) received the B.S., M.E., and Ph.D. degrees from Harbin Institute of Technology in 2004, 2007, and 2011, respectively. He is currently a Full Professor with the Department of Electronics and Communication Engineering, Harbin Institute of Technology. His research interests include wireless communications security, satellite IoT, and the integrated satellite-terrestrial communication networks. He is a Senior Member of the IEEE Communication Society, the Chair of the IEEE BTS Chapter, and the Vice Chair

of the IEEE Harbin ComSoc Chapter and the IEEE Harbin VTS Chapter. He is an Associate Editor of *IEEE CHINA COMMUNICATIONS*, *Journal of Communications and Information Networks*, *Journal of Telemetry, Tracking and Command*, and *Journal of Signal Processing*.



Zhiqiang Li (Graduate Student Member, IEEE) received the B.S. degree from the College of Information and Communication Engineering, Harbin Engineering University, Harbin, China, in 2018, and the M.E. degree in information and communication engineering from Harbin Institute of Technology (HIT), Harbin, in 2020, where he is currently pursuing the Ph.D. degree. His research interests include wireless communication and networking technologies, satellite communication, and physical layer security.



Weixiao Meng (Senior Member, IEEE) received the B.E., M.E., and Ph.D. degrees from Harbin Institute of Technology (HIT), Harbin, China, in 1990, 1995, and 2000, respectively. He worked with NTT DoCoMo, as a Visiting Researcher, from 1998 to 1999; and the University of California at Riverside, as a Senior Visiting Scholar, in 2017. He is currently a Full Professor with the School of Electronics and Information Engineering, HIT. He is a fellow of China Institute of Electronics and a Senior Member of the IEEE ComSoc and China Institute of Communication. He is the Chair of the IEEE ComSoc Harbin Chapter. He acted as the leading TPC Co-Chair of ChinaCom 2011 and ChinaCom 2016, the Awards Co-Chair of ICC 2015, the Wireless Networking Symposia Co-Chair of Globecom 2015, the AHSN Symposia Co-Chair of Globecom 2018 and ICC 2020, and the leading Workshop Co-Chair of ICC 2019 and ICNC 2020. Under his leading, Harbin Chapter won the IEEE ComSoc Chapter of the Year Award, Asia-Pacific Region Chapter Achievement Award, and the Personal Member and Global Activities Contribution Award in 2018.



Cheng Li (Senior Member, IEEE) received the B.E. and M.E. degrees from Harbin Institute of Technology, Harbin, China, in 1992 and 1995, respectively, and the Ph.D. degree in electrical and computer engineering from the Memorial University of Newfoundland, St. John's, NL, Canada, in 2004. He is currently a Full Professor and the Director of the School of Engineering Science, Simon Fraser University, Vancouver, Canada. His research interests include mobile ad hoc and wireless sensor networks, wireless communications and mobile computing,

switching and routing, and broadband communication networks.